Smart Predict Studio

How to Use the Live Cloud Version

Time-Series Forecasting with Python Darts, FastAPI and React

Overview

This guide explains how to access and use the live, cloud-hosted version of **Smart Predict Studio** on Render without installing anything on your local machine.

Live Application Links

Backend API (Swagger Docs)	https://timeseries-backend-qoem.onrender.com/docs
Frontend Web Application	https://timeseries-frontend.onrender.com

Workflow at a Glance

Step	What you do	Page
1	Upload Data and Layout Inspection	/upload
2	Confirm Schema and Transform Data	
3	Forecast Configuration	/configure
4	Real-Time Model Training Tournament	/training
5	Interactive Forecast Report	/report/forecast
6	Statistical Explanation Report	/report/explanation
7	Hierarchical Segment-Wise Report	/report/segment-wise
8	Starting a Fresh Session	

Before You Start: Wake Up the Servers

Important: Cloud Cold Start Procedure
The application is hosted on Render's free tier, so the servers automatically enter sleep mode after 15 minutes of inactivity. Always follow the 2-step procedure below to avoid network timeouts.

Step 1: Open and Wake Up the Backend First

1. Open the Backend Swagger link in your browser: https://timeseries-backend-qoem.onrender.com/docs

2. Wait approximately **30 to 50 seconds** for the Render instance to boot up.

3. You will know the backend is awake when the **FastAPI Interactive Swagger UI** page loads successfully.

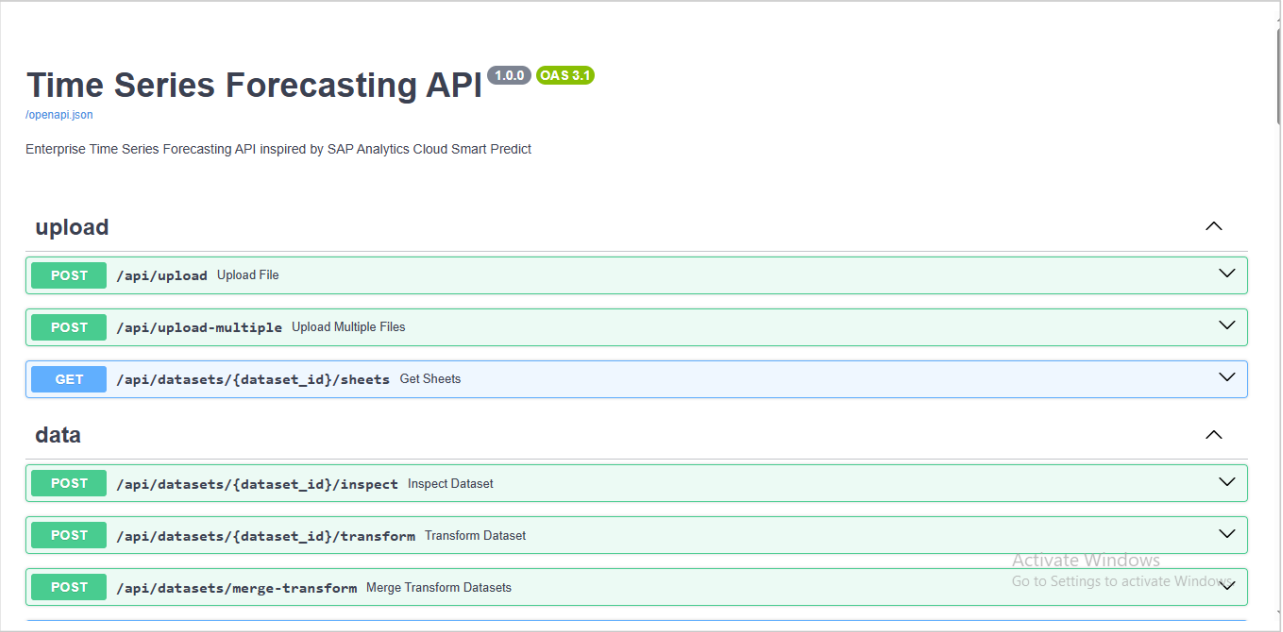


Figure 1. FastAPI Swagger UI page (backend is awake)

Step 2: Open the Frontend Application

Once the Swagger page above is visible and active:

1. Open the Frontend Application: https://timeseries-frontend.onrender.com

2. The frontend is now connected to an active backend and will communicate instantly without API timeouts or network errors.

End-to-End Forecasting Workflow (On Cloud)

Follow this step-by-step workflow in the web application.

Step 1: Upload Data and Layout Inspection (/upload)

- Navigate to the **Upload Data** tab.
- Click **Download Sample Excel** in the top action bar to download the pre-formatted reference workbook (*sample_timeseries_structure.xlsx*).
- Drag and drop a single Excel file, or switch to **Multi-File Stitching** to stitch multiple annual files (e.g. 2023, 2024, 2025).
- The system automatically detects multi-row headers (Row 1: Periods, Row 2: Dimensions).

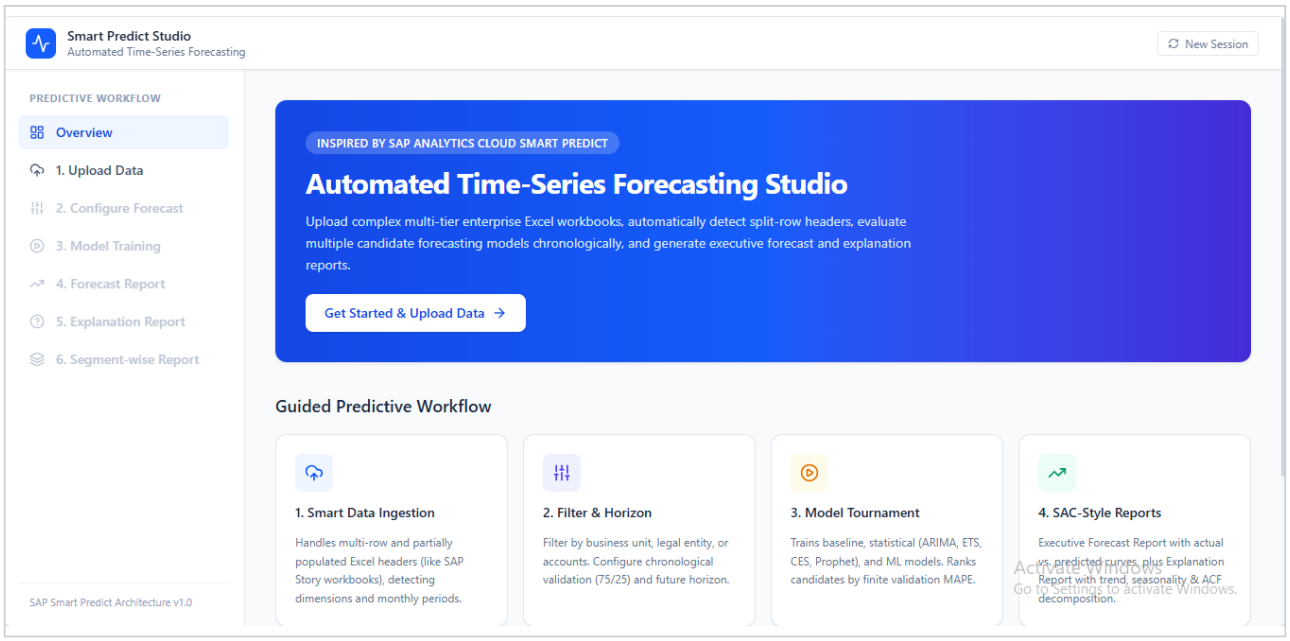


Figure 2. Data upload and multi-row header detection

Step 2: Confirm Schema and Transform Data

- Review the detected timeline start/end dates and discovered dimension columns (*Business Unit, SAC_Legal_Entity, SAC_PRODUCT, ACCOUNT, Description*).
- Adjust target account mapping if needed.
- Click **Confirm & Transform** to pivot the wide matrix into a normalized long format.
- After the transformation is complete, click **Configure Forecast**.

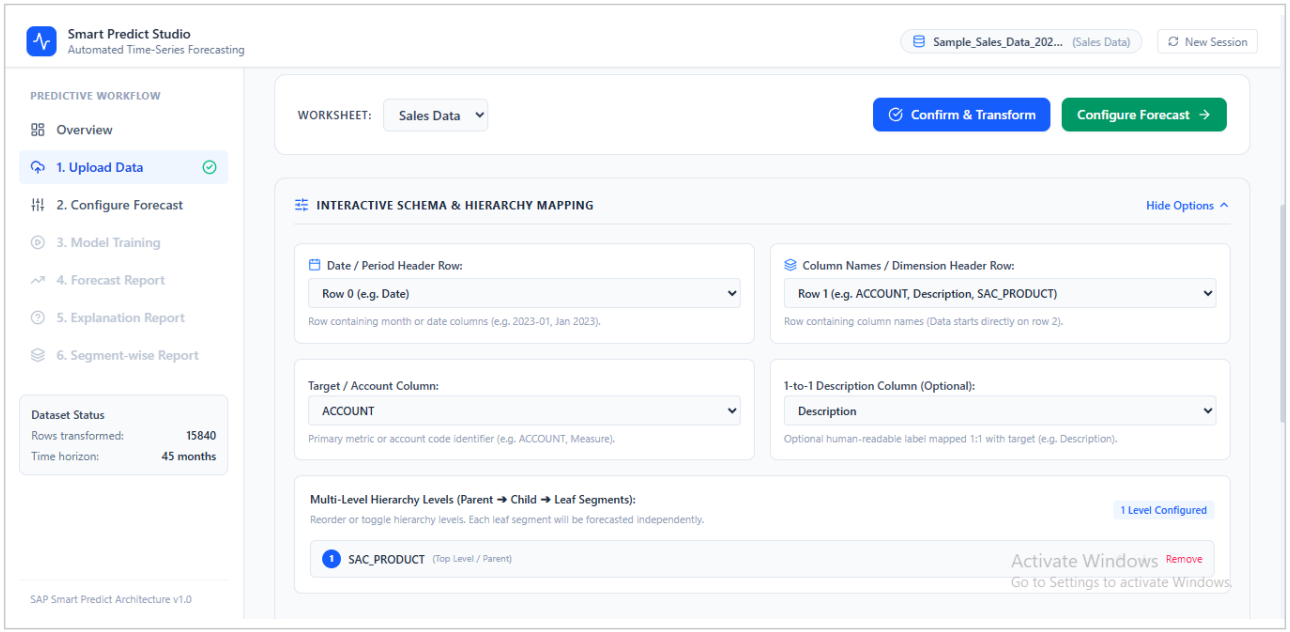


Figure 3. Confirm and transform data

Step 3: Forecast Configuration (/configure)

- **Multi-Level Hierarchy Selection (left panel):** select Business Units, Legal Entities and Product categories. Unique scoped values (e.g. "BU1-1400") prevent naming collisions between different BUs, and selected items stay pinned at the top.
- **Forecasting Target:** select your target financial or operational metric (e.g. *Sales Revenue (PL01)*).
- **Parameters and Telemetry (right panel):** the right panel has an independent scrollbar while the left panel stays locked in place. Review observation counts, select your custom training date range, and adjust the forecast horizon (defaults to the recommended 4:1 training ratio).
- Click **Train and Forecast**.

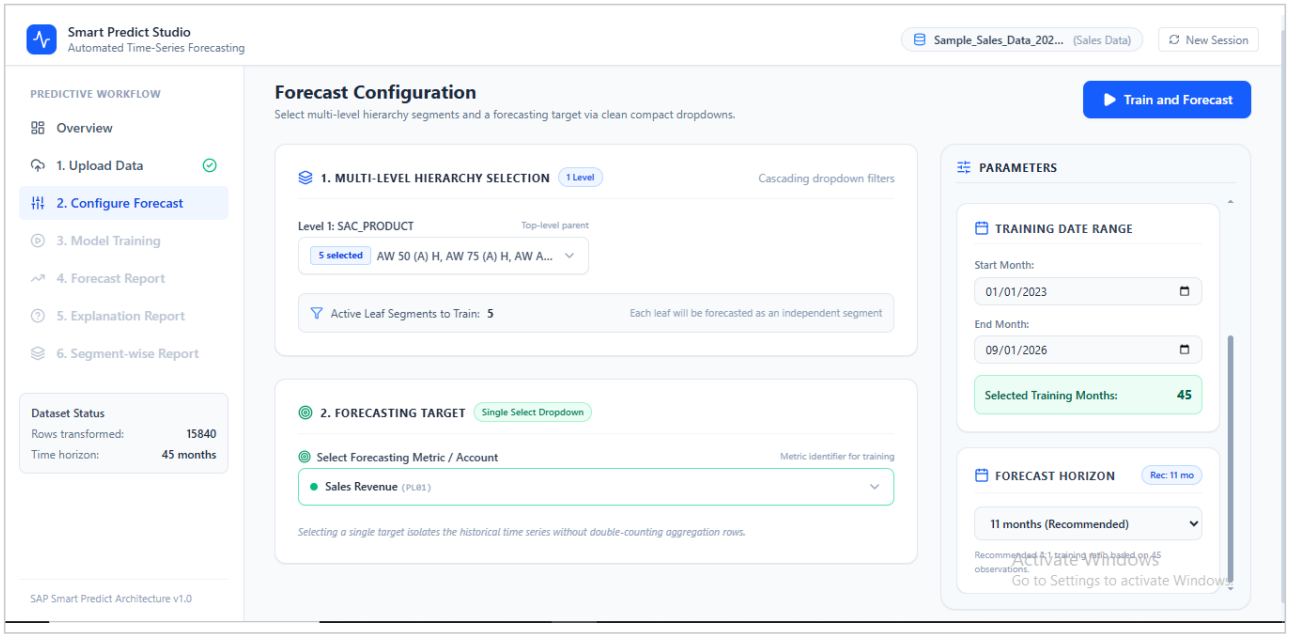


Figure 4. Forecast configuration

Step 4: Real-Time Model Training Tournament (/training)

- Watch candidate models compete side-by-side in real time across statistical (ARIMA, AutoETS, Prophet, etc.), machine learning (CatBoost, LightGBM, RandomForest, LinearRegression, XGBoost) and baseline families.
- Live progress shows real-time percentage completion, active model fitting, duration in seconds and an execution log tail.
- Once tournament evaluation completes, the winning model (lowest finite validation MAPE) is crowned champion and automatically refitted on 100% of historical observations.

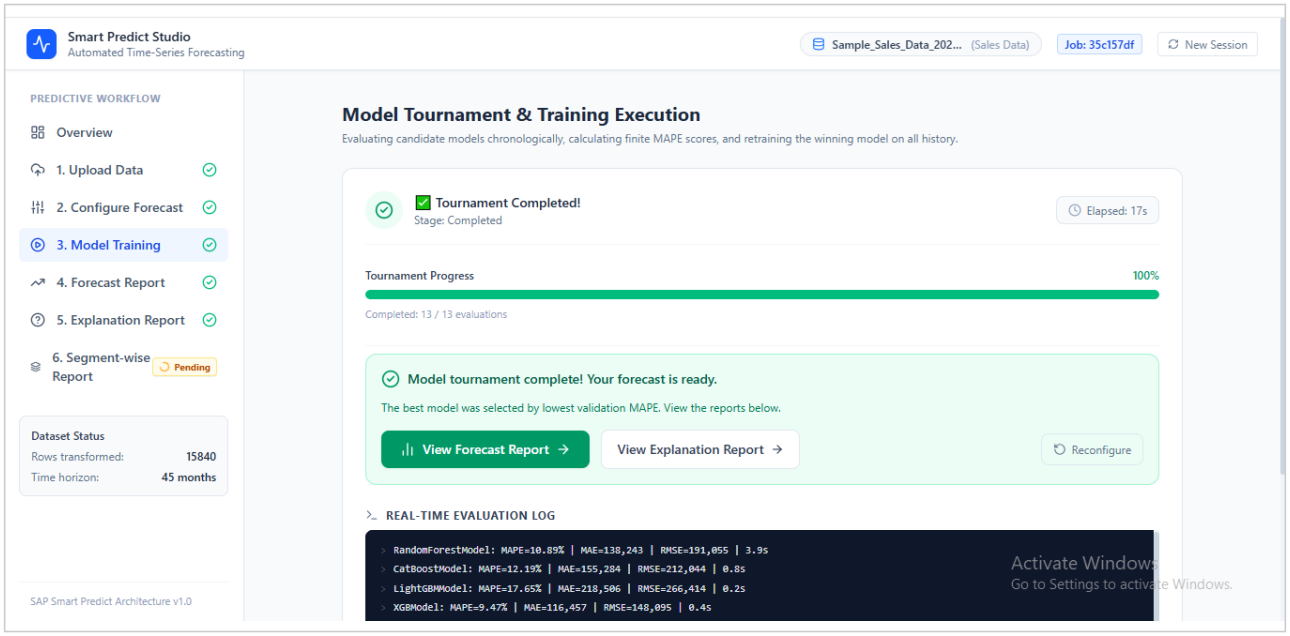


Figure 5. Model training tournament

Step 5: Interactive Forecast Report (/report/forecast)

- **Historical vs. Forecast Chart:** visualizes actual historical observations alongside the orange future forecast trajectory.
- **Validation Accuracy:** evaluates out-of-sample backtest performance (Actual vs. Predicted, MAPE, MAE, RMSE).
- **Forecast Table:** displays period-over-period forecasted figures with export options to CSV or Excel.

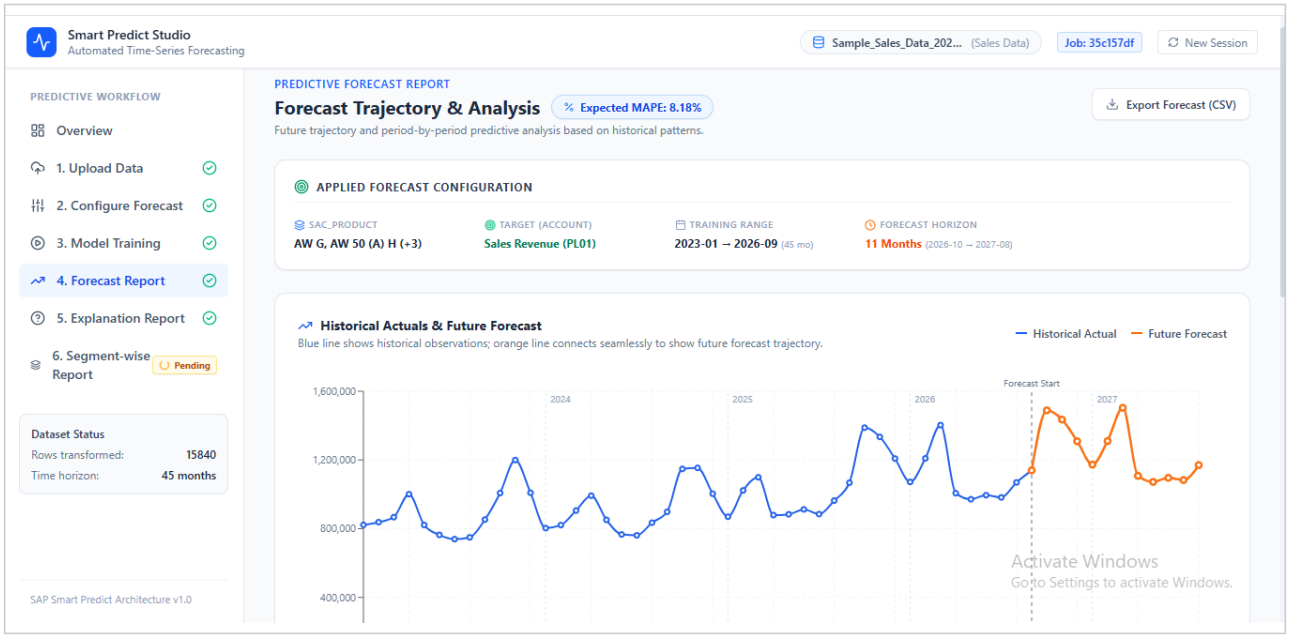


Figure 6. Historical actuals and future forecast chart

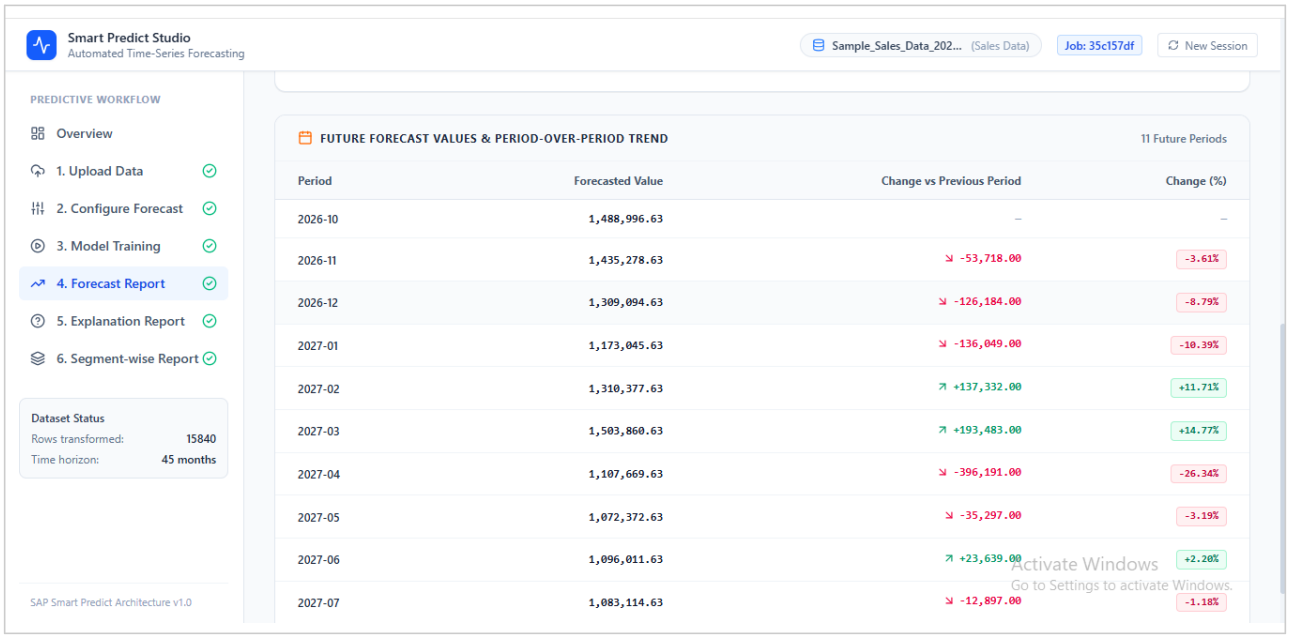


Figure 7. Forecast values and period-over-period trend

Step 6: Statistical Explanation Report (/report/explanation)

- **Trend Trajectory:** extracts linear slope direction and overall historical momentum.
- **Seasonality Pattern:** identifies recurring annual 12-month seasonal peaks and valleys.
- **Autocorrelation (ACF) Diagnostics:** plots ACF lag correlation bars to detect repeating cyclic behavior.

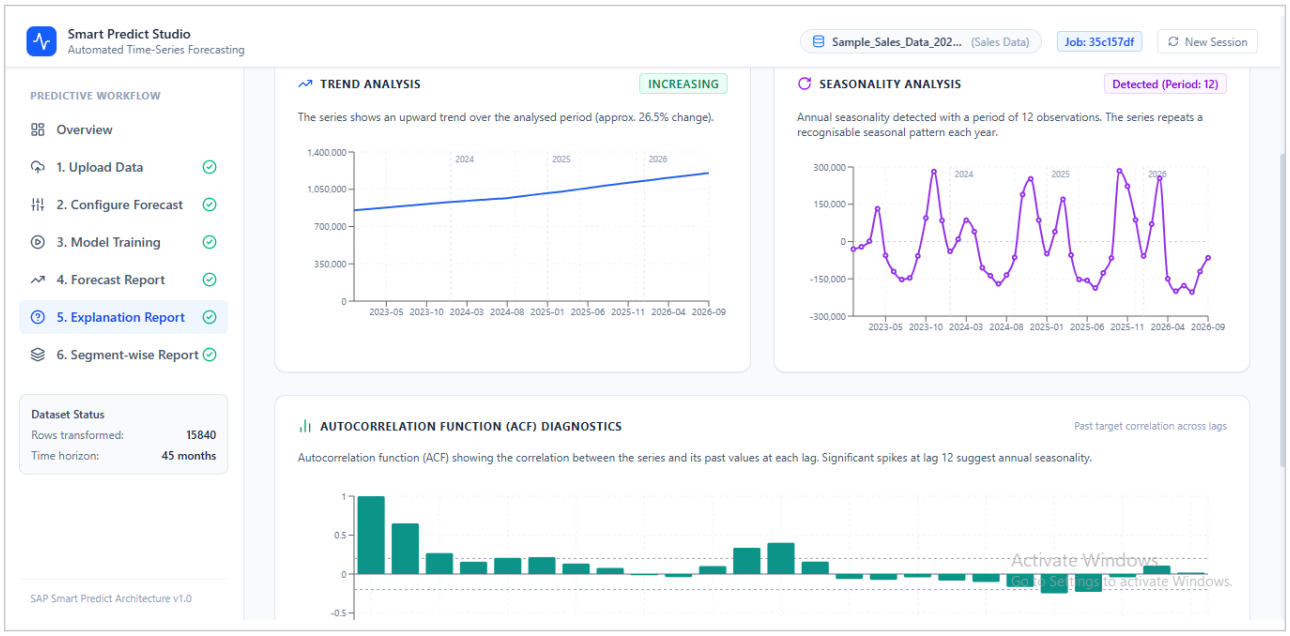


Figure 8. Statistical explanation report

Step 7: Hierarchical Segment-Wise Report (/report/segment-wise)

- Displays an independent forecast table broken down across every selected leaf segment.
- Formatted as a multi-level hierarchical matrix mirroring the SAP Analytics Cloud Story export layout.
- Provides immediate warnings for sparse segments with insufficient data (fewer than 4 points).
- Click **Export Story .xlsx** to download the complete hierarchical forecast matrix directly to Excel.

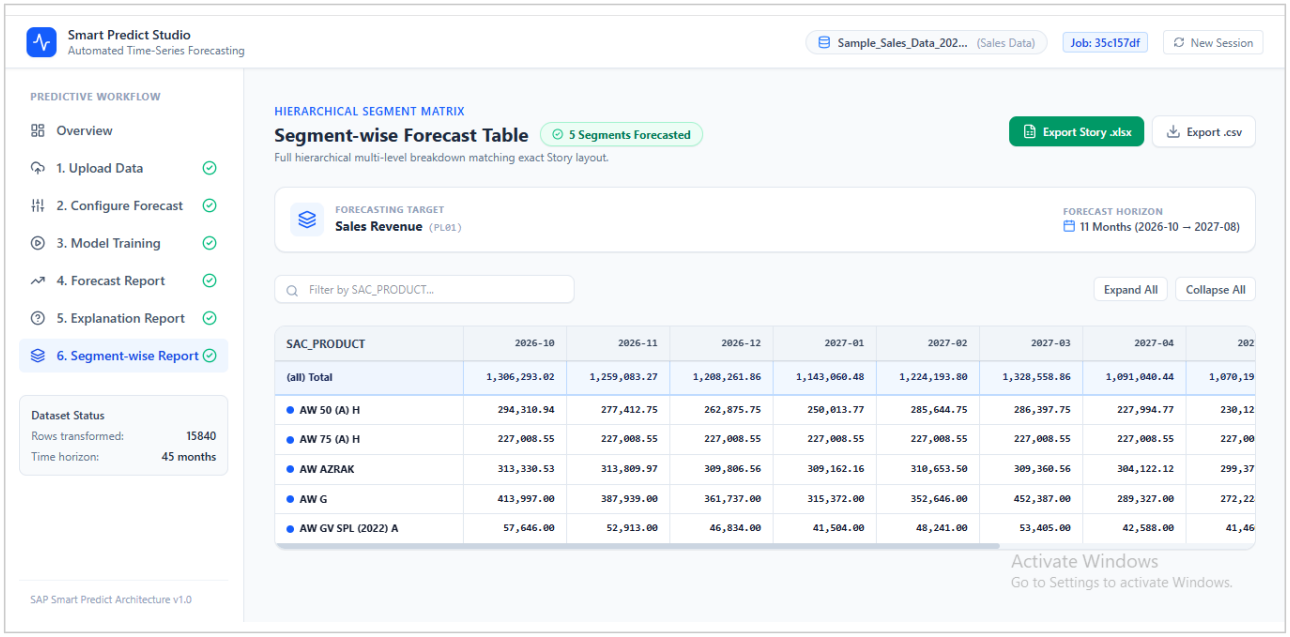


Figure 9. Segment-wise forecast table

Step 8: Starting a Fresh Session

- When you are ready to forecast a new dataset, click **New Session** in the top navigation bar, or simply upload new files on the Upload page.
- All previous tournament jobs, segment predictions and parameters are automatically reset.

Cloud Instance Notes

- **Memory Optimization:** the deployed backend incorporates low-memory streaming readers, chunked previewing, category datatype compression and garbage collection sweeps (gc.collect()) to run reliably within Render's 512 MB RAM free tier limit.